On Runs of Integers with Large Prime Factors: A Lower Bound for $k(n)$ and the Status of the Upper Bound

April 2026

Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \ldots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$k(n) \;\geq\; \exp\bigl((\log n)^{1/2-\varepsilon}\bigr)$$

for every fixed $\varepsilon > 0$ and all sufficiently large n, using Rankin's method and a pigeonhole argument.

We also discuss the upper bound $k(n) \leq \exp((\log n)^{1/2+\varepsilon})$, which would give $\log k(n) = (\log n)^{1/2+o(1)}$. The upper bound argument requires the existence of a k_1-smooth integer in every interval $[m+1, m+k_1]$ with $m \leq n$ and $k_1 = \exp((\log n)^{1/2+\varepsilon})$. We reduce this to a short-interval smooth number estimate (Hypothesis 5 below) and prove the upper bound *conditional on that hypothesis*. We discuss what the available literature (Hildebrand–Tenenbaum [1], Tenenbaum [2]) actually establishes and where the gap lies.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Set

$$\Psi(x, y) := \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. A *k-run below n* is an interval $\{m+1, \ldots, m+k\}$ with $m \le n$ and $P^+(m+j) > k$ for all $1 \le j \le k$. Define $k(n)$ to be the maximum k for which a k-run below n exists.

2 The Rankin Bound

Lemma 3 (Rankin bound [2, Theorem III.5.1])**.** *There is an effectively computable absolute constant $A \ge 3$ such that for all $x \ge y \ge 2$ with $v := \log x / \log y \ge A$,*

$$\Psi(x, y) \;\le\; x \cdot e^{-\frac{1}{2} v \log v}.$$

Proof. Cited from [2, Theorem III.5.1]. Rankin's method with the choice $\sigma = 1 - (\log v)/\log y$ yields $\Psi(x, y) \le x\, e^{-v \log v + C_0 v/\log v}$ for an absolute constant C_0. The correction term $C_0 v/\log v \le \frac{1}{2} v \log v$ once $\log^2 v \ge 2C_0$, which holds for all $v \ge A$ with $A = e^{\sqrt{2C_0}}$. The full proof with all constants is in [2, Theorem III.5.1]. $\square$

3 The Lower Bound

Theorem 4 (Lower bound)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large n, $k(n) \ge \exp\big((\log n)^{1/2-\varepsilon}\big)$.*

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: k_0-smooth numbers are sparse in $[1, n]$. Set $v_0 := \log n / \log k_0$. Since $\log k_0 = (\log n)^{1/2-\varepsilon}(1 + o(1))$, we get $v_0 = (\log n)^{1/2+\varepsilon}(1 + o(1)) \to \infty$. In particular $v_0 \ge A$ (the constant from Lemma 3) for all large n, so

$$\Psi(n, k_0) \;\le\; n \cdot e^{-\frac{1}{2} v_0 \log v_0}.$$

We estimate the exponent:

$$\tfrac{1}{2} v_0 \log v_0 = \tfrac{1}{2}(\log n)^{1/2+\varepsilon}(1+o(1)) \cdot (1/2+\varepsilon) \log\log n\, (1+o(1)) \;\ge\; (\log n)^{1/2+\varepsilon/2}$$

for all large n. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$, this gives

$$e^{-\frac{1}{2} v_0 \log v_0} \le e^{-(\log n)^{1/2+\varepsilon/2}} \ll n^{-1} \cdot k_0^{-1},$$

so in particular $\Psi(n, k_0) \le n/(2k_0)$ for all sufficiently large n.

Step 2: Pigeonhole gives an empty block. Let $B := \lfloor n/k_0 \rfloor$. Partition $\{1, \ldots, Bk_0\}$ into B blocks $I_i := [(i-1)k_0 + 1, ik_0]$. Every k_0-smooth integer in $\{1, \ldots, n\}$ lies in one of these blocks, and

$$\Psi(n, k_0) \;\le\; \frac{n}{2k_0} \;\le\; \frac{B+1}{2} \;<\; B$$

for all $n \ge 2k_0$ (i.e. all large n). By pigeonhole, some block I_i contains no k_0-smooth integer.

Step 3: The empty block is a k_0-run. For that i, every integer in I_i has largest prime factor $> k_0$. Setting $m = (i-1)k_0 \le n$, the interval $\{m+1, \ldots, m+k_0\}$ is a k_0-run below n, so $k(n) \ge k_0 = \exp((\log n)^{1/2-\varepsilon}(1 + o(1)))$. □

4 The Upper Bound: Reduction and Gap

What needs to be proved

Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. The upper bound $k(n) \le k_1$ is equivalent to: every interval $[m+1, m+k_1]$ with $m \le n$ contains at least one k_1-smooth integer.

For $m < k_1$ this is immediate: $k_1 \in [m+1, m+k_1]$ and k_1 is k_1-smooth.

For $m \in [k_1, n]$, we need the following estimate.

Hypothesis 5 (Short-interval smooth numbers)**.** Let ρ denote the Dickman function and let $\varepsilon > 0$ be fixed. There exists $n_* = n_*(\varepsilon)$ such that for all $n \ge n_*$, all $m \in [k_1, n]$ with $k_1 = \exp((\log n)^{1/2+\varepsilon})$,

$$\Psi(m + k_1,\, k_1) \;-\; \Psi(m,\, k_1) \;\ge\; 1. \qquad (1)$$

Theorem 6 (Upper bound, conditional)**.** *Assume Hypothesis 5. For every fixed $\varepsilon > 0$ and all sufficiently large n, $k(n) \le \exp((\log n)^{1/2+\varepsilon})$.*

Proof. With Hypothesis 5 every interval $[m+1, m+k_1]$ with $m \in [k_1, n]$ contains a k_1-smooth integer, and the case $m < k_1$ was handled above. Hence no k_1-run below n exists, so $k(n) < k_1$. □

Discussion: what the literature provides and where the gap lies

The main result of Hildebrand–Tenenbaum [1] is a uniform asymptotic for the *global count*. Writing $\mathcal{R}_\varepsilon = \{(x,y) : x \geq y \geq 2,\ \log y \geq (\log\log x)^{5/3+\varepsilon}\}$ and $\eta(x,y) := \Psi(x,y)/(x\rho(u)) - 1$ with $u = \log x/\log y$, the result [1, Théorème 1] states

$$\eta(x,y) \to 0 \quad \text{uniformly for } (x,y) \in \mathcal{R}_\varepsilon \text{ as } y \to \infty. \tag{2}$$

In other words: for every $\delta > 0$ there exists $Y_0(\varepsilon,\delta)$ such that $|\eta(x,y)| \leq \delta$ for all $(x,y) \in \mathcal{R}_\varepsilon$ with $y \geq Y_0$. This is a statement about y alone: Y_0 does *not* depend on x.

The subtraction approach and why it fails here

Subtracting (2) at $x+y$ and at x:

$$\Psi(x+y,y) - \Psi(x,y) \;=\; \underbrace{(x+y)\rho(u') - x\rho(u_1)}_{\text{(I)}} \;+\; \underbrace{(x+y)\rho(u')\eta_1 - x\rho(u_1)\eta_2}_{\text{(II)}}, \tag{3}$$

where $u' = \log(x+y)/\log y$, $u_1 = u$, $\eta_j = \eta(\cdot)$ evaluated at the respective arguments.

Term (I). A mean-value argument using $\rho'(t) = -\rho(t-1)/t$ and $u' - u_1 = \log(1+y/x)/\log y \leq (y/x)/\log y$ gives

$$\text{(I)} \;=\; y\rho(u_1)\left(1 + O\left(\frac{1}{\log y}\right)\right), \tag{4}$$

valid for $x \geq y$ and $y \geq 2$. (The calculation: expand $g(t) = t\rho(\log t/\log y)$ by the mean-value theorem; the derivative $g'(t) = \rho(u) - \rho(u-1)/(u\log y)$ has a relative correction $O(1/\log y)$ since $\rho(u-1)/\rho(u) = O(u)$ and $u \leq u_1 \leq (\log n)^{1/2-\varepsilon} = o(\log y)$.) So Term (I) is the main term, equal to $k_1\rho(u_1)(1 + O(1/\log k_1))$ with a *relative* error $O(1/\log k_1) \to 0$.

Term (II) is the obstacle. With $|\eta_1|, |\eta_2| \leq \delta$:

$$|\text{(II)}| \;\leq\; \big[(x+y)\rho(u') + x\rho(u_1)\big]\delta \;\approx\; 2x\rho(u_1)\delta.$$

In our application $x = m$ and $y = k_1$, so $|\text{(II)}| \lesssim 2m\rho(u_1)\delta$. The ratio of this error to the main term is

$$\frac{|\text{(II)}|}{y\rho(u_1)} \;\approx\; \frac{2m\delta}{k_1}.$$

For $m \in [k_1, n]$ this ratio ranges up to $2n\delta/k_1$. Now $n/k_1 = \exp(\log n - (\log n)^{1/2+\varepsilon}) \to \infty$, so *no matter how small* δ *is* (as long as it is a fixed positive number, even an astronomically small one), the ratio $n\delta/k_1 \to \infty$. The error term dominates the main term for large m.

Conclusion: the naive subtraction of two applications of (2) does *not* prove Hypothesis 5. The "$|\eta| \le \delta$ uniformly in x" form of the HT theorem is insufficient because two near-cancelling large quantities are being subtracted.

What would close the gap: two routes

Route 1: A Lipschitz bound on the error term. If the HT error satisfies, for $(x, y) \in \mathcal{R}_\varepsilon$ and $y \ge Y_1(\varepsilon)$,

$$|\eta(x + y, y) - \eta(x, y)| \;\le\; \frac{C\,y}{x \log y} \tag{5}$$

for some absolute constant C, then in (3):

$$|(\mathrm{II})| \le (x + y)\rho(u')|\eta_1 - \eta_2| + O(y\rho(u_1)|\eta_2|) \le \frac{2Cy\rho(u_1)}{\log y} + O(y\rho(u_1)\delta).$$

Choosing $\delta = \frac{1}{4}$ and y large enough so $2C/\log y \le \frac{1}{4}$, Term (II) is at most $\frac{1}{2}y\rho(u_1)$, which is half the main term. Combined with $k_1\rho(u_1) \ge 2$ (established above), the count is ≥ 1.

Bound (5) says: as x varies by one step of size y, the *relative error* η changes by at most $Cy/(x \log y)$, i.e. the relative error is Lipschitz in x with a Lipschitz constant $O(1/(x \log y))$. This is the kind of equicontinuity that one expects from the saddle-point analysis (the saddle-point location σ_0 shifts smoothly with x), but it does not appear explicitly in [1, 2].

Route 2: Direct Perron analysis of the short-interval integral. Rather than subtracting two global estimates, one can analyze the Perron integral

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) = \frac{1}{2\pi i}\int_{\sigma_0 - iT}^{\sigma_0 + iT} F_{k_1}(s)\,\frac{(m + k_1)^s - m^s}{s}\,ds + (\text{tail}) \tag{6}$$

directly, where $F_y(s) = \prod_{p \le y}(1 - p^{-s})^{-1}$ and $\sigma_0 = 1 - (\log u_1)/\log k_1$ is the saddle point. The kernel satisfies

$$(m + k_1)^s - m^s = m^s\big[(1 + k_1/m)^s - 1\big] = m^s\Big[s\,\frac{k_1}{m} + O\Big(s^2\,\frac{k_1^2}{m^2}\Big)\Big].$$

The leading term $s\,k_1 m^{s-1}$ is, up to the factor s, the integrand for $\Psi'(m, k_1)\cdot k_1$ (the "derivative" of Ψ with respect to m). The HT saddle-point analysis of $F_{k_1}(s)m^{s-1}$ at $s = \sigma_0$ gives the main term $m^{-1} \cdot m\rho(u_1) = \rho(u_1)$, so the leading contribution to (6) is $k_1\rho(u_1)(1 + O(1/\log k_1))$.

The correction from the $O(s^2k_1^2/m^2)$ term contributes $O(k_1^2/m^2)\cdot\frac{1}{2\pi i}\int F_{k_1}(s)m^s s\,ds$. The integral $\frac{1}{2\pi i}\int F_{k_1}(s)m^s s\,ds$ is of order $m\rho(u_1)\sigma_0 \log m \le m\rho(u_1)\log n$ (bounded by differentiating the Perron formula for $\Psi(m, k_1)$). Hence the correction is $O(k_1^2\rho(u_1)\log n/m) = O(k_1\rho(u_1) \cdot k_1 \log n/m)$, which is $o(k_1\rho(u_1))$ whenever $m \gg k_1 \log n$.

For $m \le Ck_1 \log n$ (a thin initial range), the interval $[m + 1, m + k_1]$ contains $m + k_1 \le (C \log n + 1)k_1$, so the relevant multiple of k_1 in that interval is jk_1 with $j \le C \log n + 1$. For $j \le k_1$, jk_1 is k_1-smooth (since all prime factors of j are $\le j \le k_1$). As $k_1 = e^{(\log n)^{1/2+\varepsilon}} \gg \log n$, the condition $j \le k_1$ is satisfied throughout this initial range.

Thus Route 2 would yield Hypothesis 5 by splitting into $m \le Ck_1 \log n$ (where $jk_1 \in [m + 1, m + k_1]$ with $j \le k_1$ is a k_1-smooth witness) and $m > Ck_1 \log n$ (where the direct Perron analysis gives $k_1\rho(u_1)(1+o(1)) \ge 1$). The second part requires completing the saddle-point analysis of (6) uniformly in m, with all error terms made explicit. This is within the scope of the HT method but has not been written out in this form in [1, 2].

Remark 7 (What an unconditional proof would require)**.** Both routes above are "incrementally hard" in the sense that they require tightening known results rather than new ideas. Concretely:

- Route 1 requires the HT saddle-point method to yield a second-order expansion $\Psi(x, y) = x\rho(u)[1 + c_1(u)/\log y + O((\log y)^{-2})]$ with $c_1(u)$ uniformly Lipschitz in u, together with a statement that this expansion is uniform in x (not just for fixed x).

- Route 2 requires the direct Perron analysis of the short-interval integral (6) to be carried through to completion, tracking saddle-point and tail errors uniformly in $m \in [Ck_1 \log n, n]$.

Either approach is a refinement of existing technology. The key inputs (saddle-point for $F_y(s)m^s$, bounds on the Dickman function and its derivative, Mertens-type estimates for $\prod_{p\le y}(1 - p^{-\sigma})^{-1}$) are all available in [1, 2].

5 Main Result

Theorem 8. (a) *(Unconditional) For every fixed* $\varepsilon > 0$ *and all large* n*:* $k(n) \geq \exp((\log n)^{1/2-\varepsilon})$.

(b) *(Conditional on Hypothesis 5) For every fixed* $\varepsilon > 0$ *and all large* n*:* $k(n) \leq \exp((\log n)^{1/2+\varepsilon})$.

Hence, conditional on Hypothesis 5, $\log k(n) = (\log n)^{1/2+o(1)}$*. Hypothesis 5 is a short-interval smooth number estimate plausibly within reach of the Hildebrand–Tenenbaum method but not verified directly from the published sources [1, 2] here.*

Proof. Part (a) is Theorem 4. Part (b) is Theorem 6. □

References

[1] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.

[2] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 3rd ed., 2015.